

EclipsePhoto

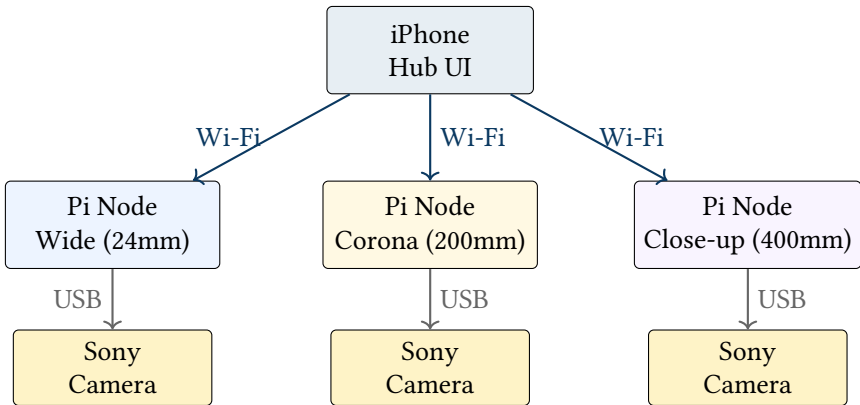
Autonomous Edge Control for Total Solar Eclipse Capture

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EclipsePhoto Project

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Abstract. EclipsePhoto is an autonomous eclipse photography control system designed to capture total solar eclipse events across their full dynamic and temporal range. Built on Raspberry Pi capture nodes coordinated by an iPhone hub, the system manages the critical challenges unique to eclipse photography: a 17 EV illuminance swing in minutes, safety-critical solar filter management, HDR bracket sequences spanning 12+ EV stops of corona brightness, and sub-second transient phenomena (diamond ring, chromosphere flash). This report details the astronomical science, optical engineering, and operational procedures that inform the system design, from eclipse geometry and contact-point timing through multi-rig deployment strategy and real-time telemetry.

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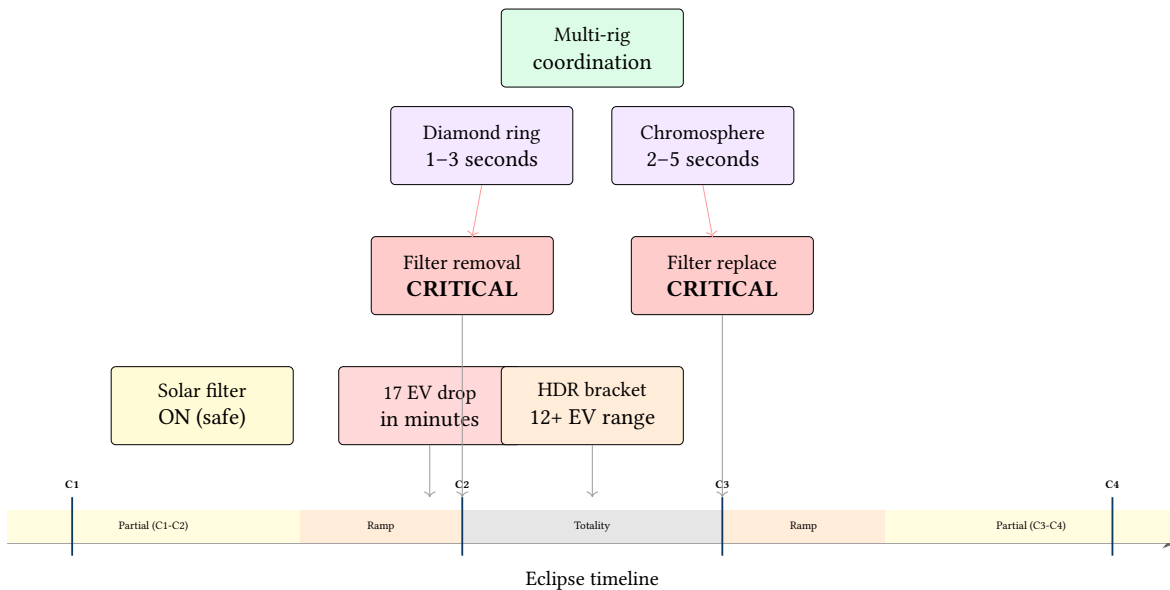
1 Problem Framing

Total solar eclipse photography presents a uniquely adversarial control problem. Unlike aurora photography—where displays recur and the photographer can iterate over multiple nights—a total eclipse at a given location is a once-in-a-lifetime event. The path of totality is roughly 150 km wide and sweeps across the Earth’s surface at over 1 500 km/h; at any given site, totality lasts between 2 and 7 minutes. There are no second chances.

Within those few minutes, the photographer must contend with five interlocking challenges:

1. **Extreme illuminance range.** Ambient light drops from ~100,000 lux (full sunlight) to ~1 lux (deep twilight equivalent) in the final minutes before C2, a span of approximately 17 EV stops. The exposure ramp must track this change on a cadence of seconds, not minutes.
2. **Safety-critical solar filter management.** A solar ND filter (optical density ~5, transmitting ~0.001% of incident light) must be in place during all partial phases to protect both the camera sensor and the operator’s eyes. Forgetting to *remove* the filter at C2 wastes the entire totality window. Forgetting to *replace* it at C3 risks immediate sensor damage from unfiltered sunlight.

3. **Sub-second transient phenomena.** The diamond ring effect (a single brilliant point of sunlight shining through a lunar valley) lasts 1–3 seconds. The chromosphere flash (a thin red ring of hydrogen emission) is visible for 2–5 seconds at C2 and again at C3. Both require pre-armed burst capture at the correct exposure—there is no time to meter and adjust.
4. **Multi-rig coordination.** A serious eclipse deployment involves two or three camera rigs at different focal lengths (wide-angle for context, telephoto for corona, long telephoto for prominences). Each rig requires synchronized filter management, exposure ramping, and bracket sequencing.
5. **Operator cognitive overload.** Totality triggers a well-documented “awe response” that degrades the operator’s ability to perform precise manual tasks. Experienced eclipse photographers report forgetting entire steps of their carefully rehearsed plan. Automation is not merely convenient—it is necessary for reliable capture.



Insight:
The key product value of EclipsePhoto is not intervalometer-style automation but rather time-critical orchestration: managing filter state, exposure ramps, and bracket sequences across multiple rigs during a window where human cognition is unreliable.

Figure 1. Eclipse photography challenge overview: safety-critical filter management at C2/C3, a 17 EV illuminance swing requiring rapid exposure adaptation, sub-second transient events (diamond ring, chromosphere), and the need for coordinated multi-rig control—all during a window where operator cognition is degraded by the awe response.

2 Eclipse Geometry and Contact Points

A total solar eclipse progresses through four contact points (C1–C4) as the Moon traverses the solar disc. The system must track these stages because illuminance changes by a factor of $\sim 10,000\times$ between partial phases and totality, requiring radically different exposure parameters at each transition.

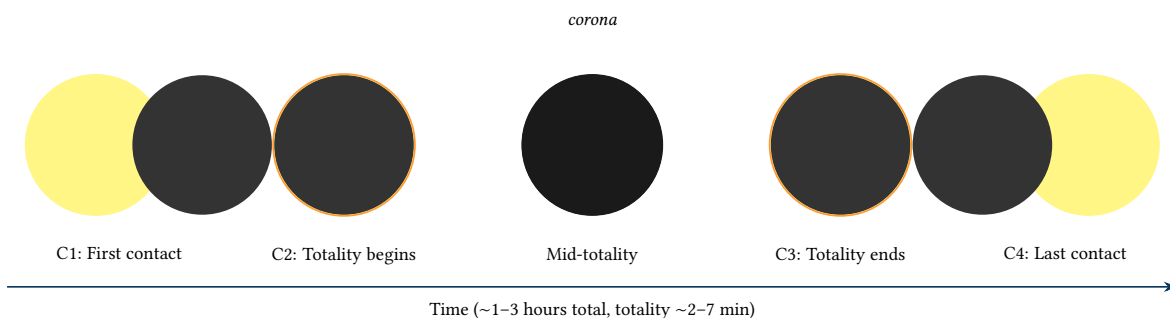


Figure 2. Solar eclipse contact sequence: C1 (first external tangency), C2 (totality onset), mid-totality with corona visible, C3 (totality end), C4 (last external tangency).

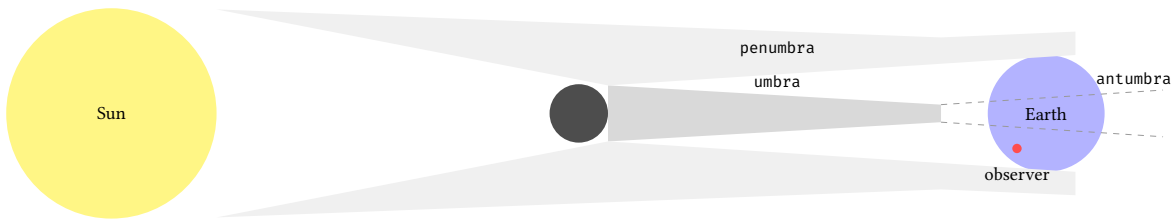


Figure 3. Eclipse shadow geometry (cross-section): the Moon casts an umbral cone (total shadow), penumbral cone (partial shadow), and antumbra (annular shadow region beyond the umbral vertex).

Each contact point has specific photographic implications:

- **C1 (First Contact):** The Moon's limb first touches the solar disc. Begin partial-phase capture with the solar filter in place. A cadence of 1 frame every 2 minutes is sufficient to document the Moon's progression across the Sun. Sunspot transit can be captured if active regions are present.
- **C2 (Second Contact):** The Moon fully covers the solar disc and totality begins. This is the single most critical transition: the solar filter *must* be removed, and HDR bracket capture must begin immediately. The diamond ring effect occurs in the final 1–3 seconds before C2, requiring burst capture at fast shutter speeds (1/2000–1/4000s). The chromosphere flash (thin red H α ring) is visible for 2–5 seconds around C2.
- **C3 (Third Contact):** Totality ends as the first sliver of photosphere reappears. The diamond ring and chromosphere flash occur again in the reverse sequence. Immediately after capturing the C3 diamond ring, the solar filter *must* be replaced to protect the sensor from the rapidly increasing sunlight.
- **C4 (Fourth Contact):** The Moon's limb fully separates from the solar disc. End of partial-phase capture. Stop the intervalometer sequence.

3 Light Levels Across Eclipse Stages

The illuminance change during a total solar eclipse spans roughly five orders of magnitude. Understanding this curve is essential for designing the exposure ramp strategy: the system must transition smoothly from bright-sun settings to near-nighttime settings and back within minutes.

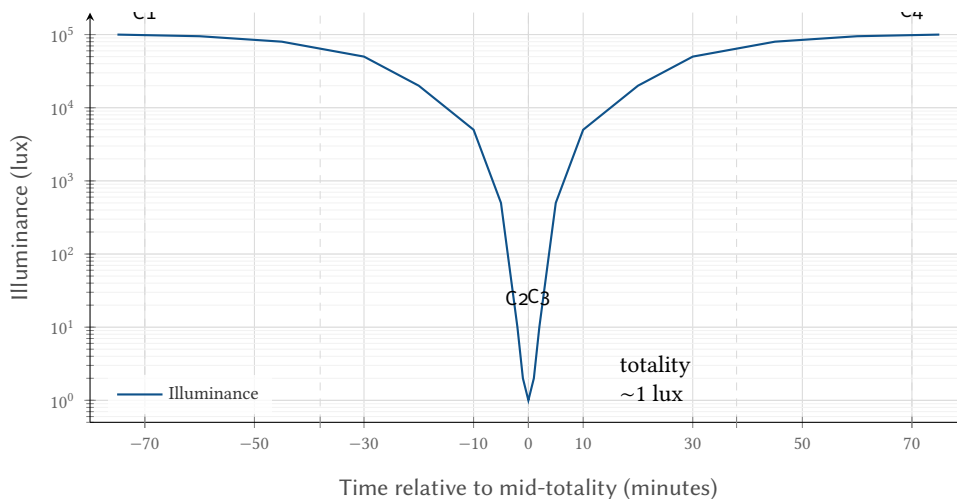


Figure 4. Solar eclipse illuminance curve (log scale). The drop from ~100,000 lux to ~1 lux occurs over just a few minutes around C2/C3, requiring a ~17 EV stop adjustment.

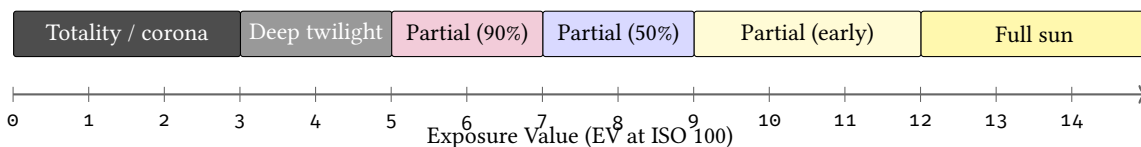


Figure 5. Equivalent EV scale showing the range of scene brightness from totality (EV 0–3) through partial eclipse stages to full sunlight (EV 14–15).

Eclipse Stage	Illuminance	EV	Notes
Full sun (pre-C1)	100,000 lux	14–15	Solar filter required
Partial (25% covered)	75,000 lux	14	Solar filter still required
Partial (50% covered)	50,000 lux	13	Solar filter still required
Partial (90% covered)	10,000 lux	11	Noticeable dimming
Partial (99% covered)	1,000 lux	8	Overcast-equivalent
Diamond ring / Bailey's beads	10–100 lux	4–6	Remove solar filter
Totality (corona)	0.5–5 lux	0–3	Wide exposure bracket needed

Table 1. Typical illuminance and EV values at each solar eclipse stage.

The rate of illuminance change near C2 and C3 is highly non-linear. During most of the partial phase (C1 to roughly 10 minutes before C2), the dimming is gradual and almost imperceptible—from 100,000 lux to perhaps 20,000 lux over 50 minutes. But in the final 5 minutes before C2, illuminance plummets from ~5,000 lux to ~1 lux: a factor of 5,000× in 300 seconds. This means the exposure must react on a cadence of approximately 5 seconds during the ramp period, adjusting by 1–2 EV stops per update.

The practical consequence is a “twilight zone” of approximately 30 seconds on either side of C2/C3 where manual exposure management is simply impractical. The illuminance is changing too rapidly for a human to meter, evaluate, and adjust before the scene has shifted by another 2–3 stops. This is the window where automated exposure ramping provides its greatest value.

Non-linear Rate of Change

The illuminance curve near C2/C3 is approximately exponential, not linear. Fixed-interval exposure steps (e.g., “adjust 1 stop every 30 seconds”) are insufficient because the required adjustment rate *accelerates* as totality approaches. EclipsePhoto must implement an accelerating ramp that increases the step frequency from once per minute (early partial) to once per 3–5 seconds (final approach to C2).

4 Eclipse Stage Detail: What Happens Physically

Each eclipse contact point involves specific physical phenomena that produce distinct photographic opportunities. Understanding what is visible at each stage—and for how long—drives both the exposure plan and the frame rate required to capture transient events.

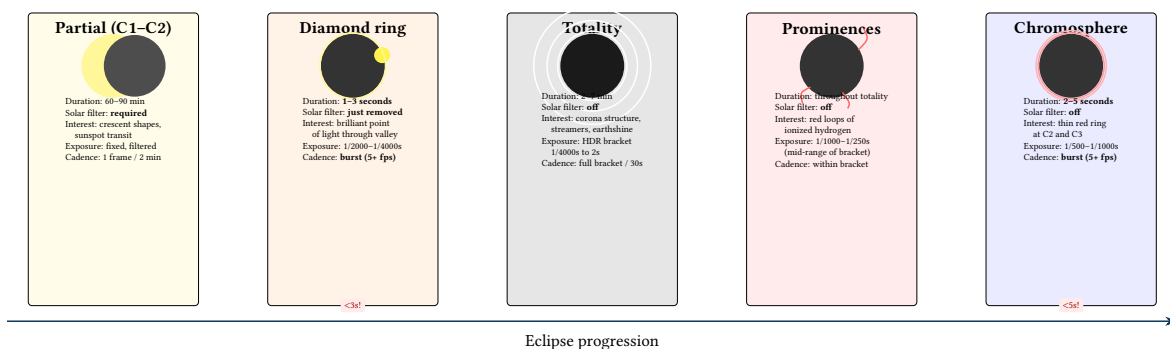


Figure 6. Eclipse stage detail: each phase presents distinct photographic targets with dramatically different durations and exposure requirements. Diamond ring and chromosphere events last only seconds and require burst-mode capture with pre-configured settings.

Transient Events

The diamond ring (1–3s) and chromosphere flash (2–5s) are the most time-critical moments of the eclipse. Missing them means waiting for the next eclipse. EclipsePhoto should pre-arm burst capture at these transition points with exposure settings locked to the expected brightness.

5 Color Temperature Progression During Eclipse

The correlated color temperature (CCT) of ambient light changes significantly during a solar eclipse. This affects camera auto white balance behavior and the appearance of both the eclipse itself and the surrounding landscape. EclipsePhoto should lock white balance rather than allowing auto adjustment.

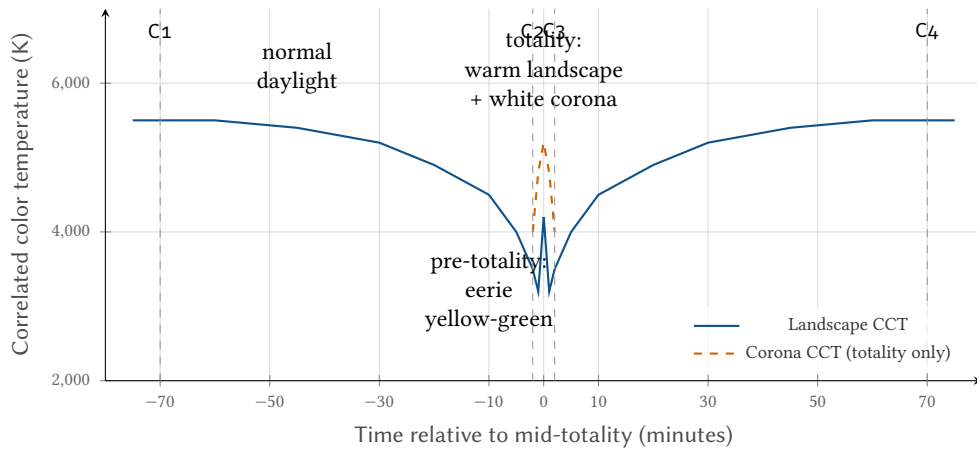


Figure 7. Color temperature progression: landscape CCT drops from ~5500 K (daylight) to ~3200 K just before totality as atmospheric scattering shifts toward warm tones. The corona itself is ~5000–5200 K (white). Auto white balance will oscillate—lock WB before the eclipse begins.

: White Balance Lock

Lock camera white balance to a fixed value (5200–5500 K daylight) before C1. Auto white balance will hunt during the rapid color temperature changes and produce inconsistent frames that complicate post-processing and timelapse assembly.

6 Dynamic Range Challenge: Corona and Prominences

The solar corona alone spans approximately 12 EV stops of brightness—far exceeding the dynamic range of any single exposure. The inner corona (within 0.5 solar radii) is roughly 1 million times brighter than the outer corona (at 3+ solar radii). Capturing the full extent requires HDR bracketing with a wide spread of exposures.

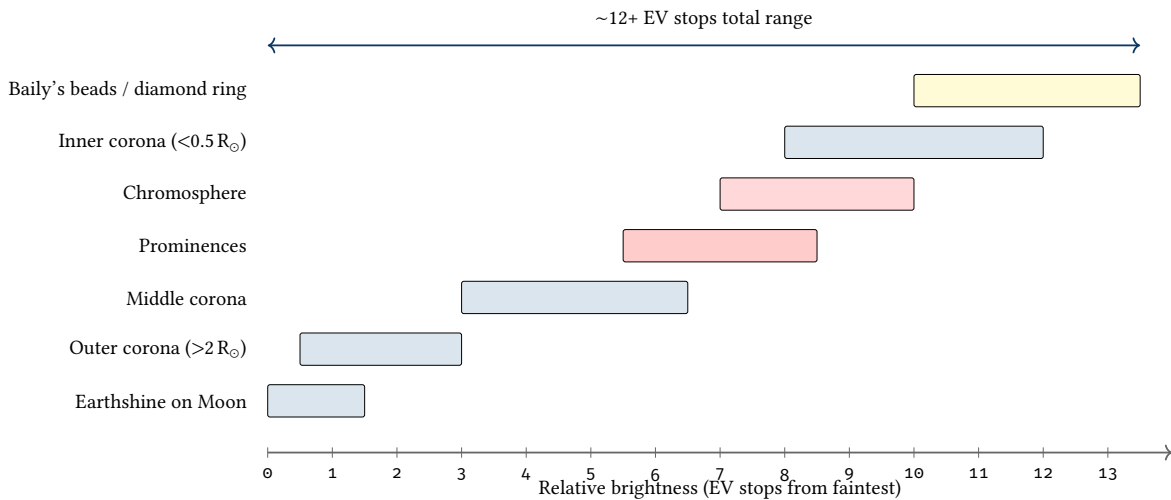


Figure 8. Dynamic range of eclipse features: from faint earthshine on the lunar surface to the brilliant diamond ring, the brightness span exceeds 12 EV stops.

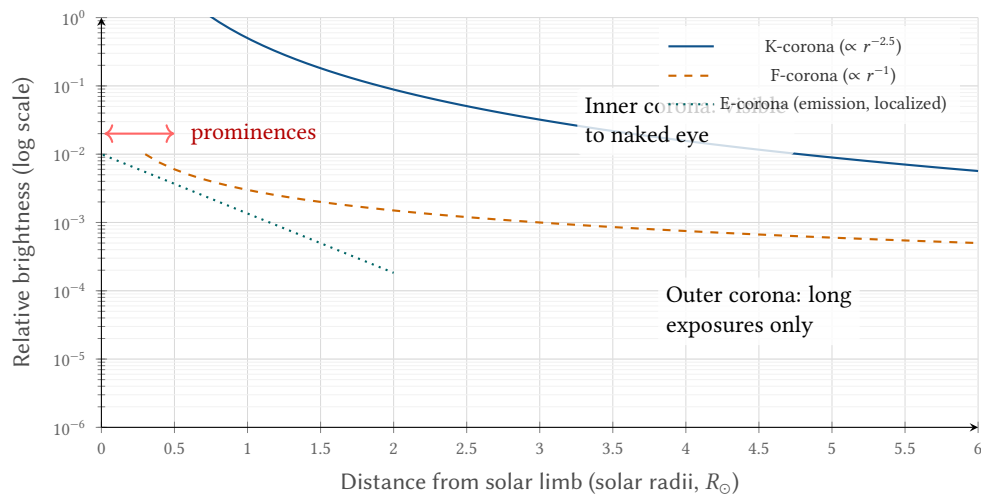


Figure 9. Corona radial brightness profile: the K-corona (electron-scattered) dominates the inner region and drops steeply ($\sim r^{-2.5}$), while the F-corona (dust-scattered) dominates at larger distances with a shallower decline. This radial gradient is the physical reason the dynamic range exceeds 12 EV stops.

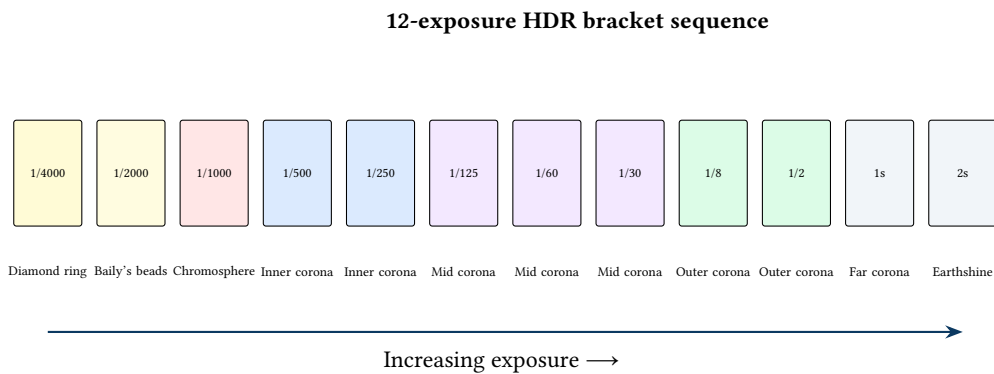


Figure 10. Bracketing strategy: a 12-exposure sequence from 1/4000s to 2s, with each exposure targeting a specific eclipse feature. The full bracket captures the complete dynamic range for HDR compositing.

6.1 Solar Activity and Prominence Monitoring

Before the eclipse, the operator should consult solar activity data from NOAA's Space Weather Prediction Center (SWPC) and NASA's Solar Dynamics Observatory (SDO) to anticipate prominence activity. Active regions near the solar limb may produce prominences visible during totality—and their size can vary dramatically, from small loops barely extending beyond the limb to exceptional eruptions reaching 0.5 solar radii or more.

Prominences are plasma structures suspended in the corona by magnetic fields. Their height above the solar limb determines whether they will be visible to the naked eye during totality (most are) and whether they will fit within the frame of a tightly cropped telephoto image (some will not).

Insight:
The corona alone spans ~12 EV stops—no single exposure captures it. Automated HDR bracketing at totality onset is the single most critical automated action.

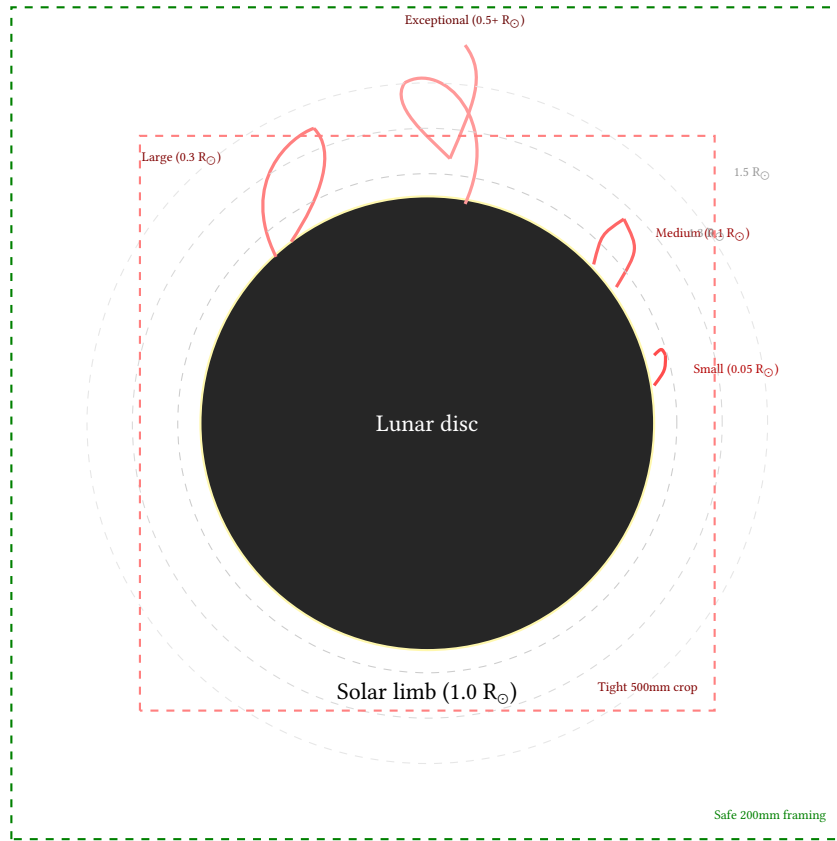


Figure 11. Prominence heights relative to the solar disc: small prominences ($0.05 R_{\odot}$) are common and fit any framing; exceptional eruptions ($0.5+ R_{\odot}$) can extend well beyond a tight telephoto crop. The red dashed box shows how a 500 mm crop risks cutting off large prominences; the green dashed box shows safer 200 mm framing.

6.2 Framing for Prominences

When composing the telephoto frame, leave generous room beyond the solar limb for prominences. A tight crop that fills the frame with the lunar disc and inner corona may produce a dramatic image of streamers, but risks missing active-region prominences that extend beyond expectations. Solar maximum periods (such as 2024–2026) produce more frequent and larger prominences, making this consideration especially important.

The safest approach is to frame at a focal length that provides at least $1.5 R_{\odot}$ of clearance beyond the limb in all directions. At 200 mm on a full-frame sensor, the solar disc occupies roughly 10% of the frame width—leaving ample room. At 400 mm, the disc fills about 20%, still reasonable. Beyond 600 mm, the risk of cropping prominences increases significantly.

7 Camera Parameter Adaptation Strategy

The extreme dynamic range of eclipse events requires a systematic approach to camera parameter changes. The strategy prioritizes shutter speed adjustment (fastest response, preserves optical quality) over ISO changes (affects noise floor), while holding aperture fixed (maintains optical performance and depth of field).

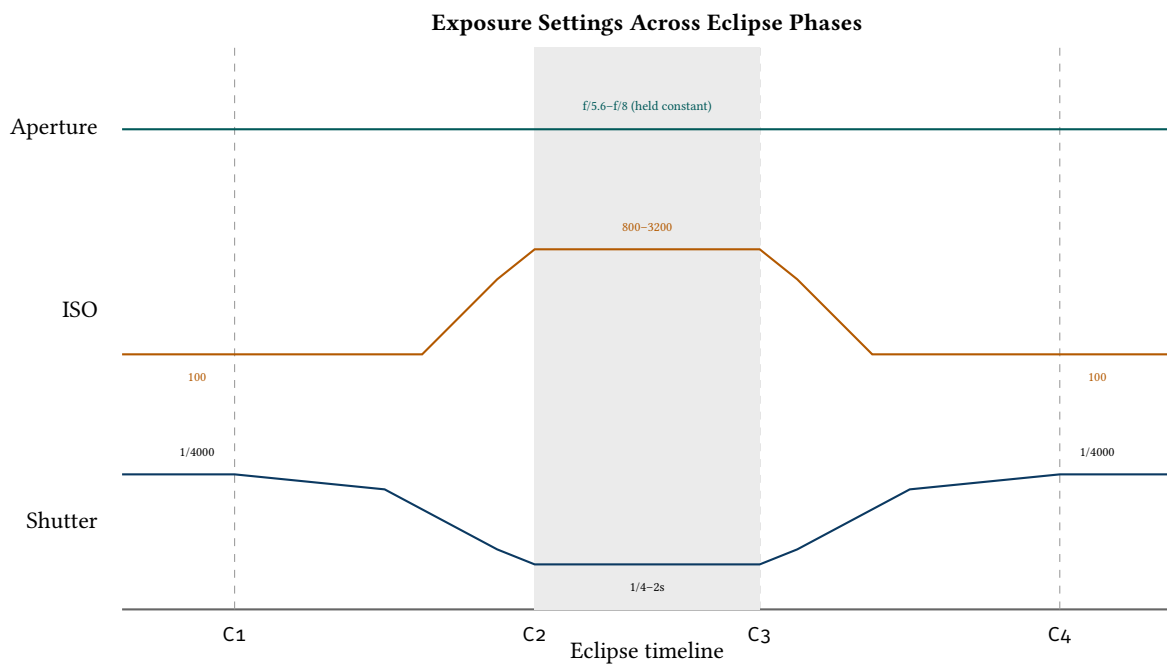


Figure 12. Multi-track exposure timeline: shutter speed and ISO ramp dramatically around C2/C3 while aperture is held constant to maintain optical quality.

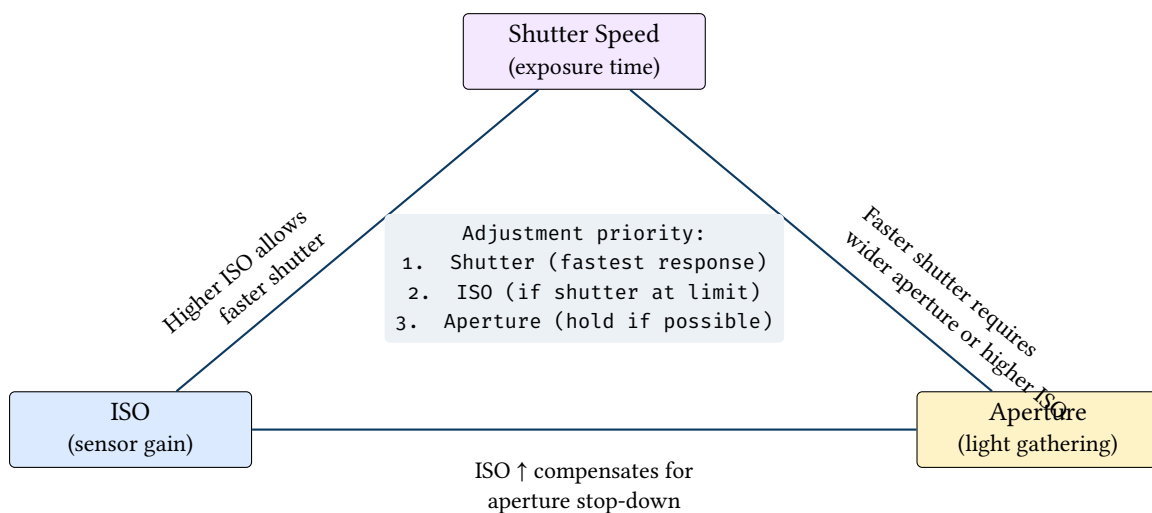


Figure 13. Exposure triangle: the three interdependent parameters. EclipsePhoto’s control strategy prioritizes shutter adjustment, then ISO, while holding aperture fixed.

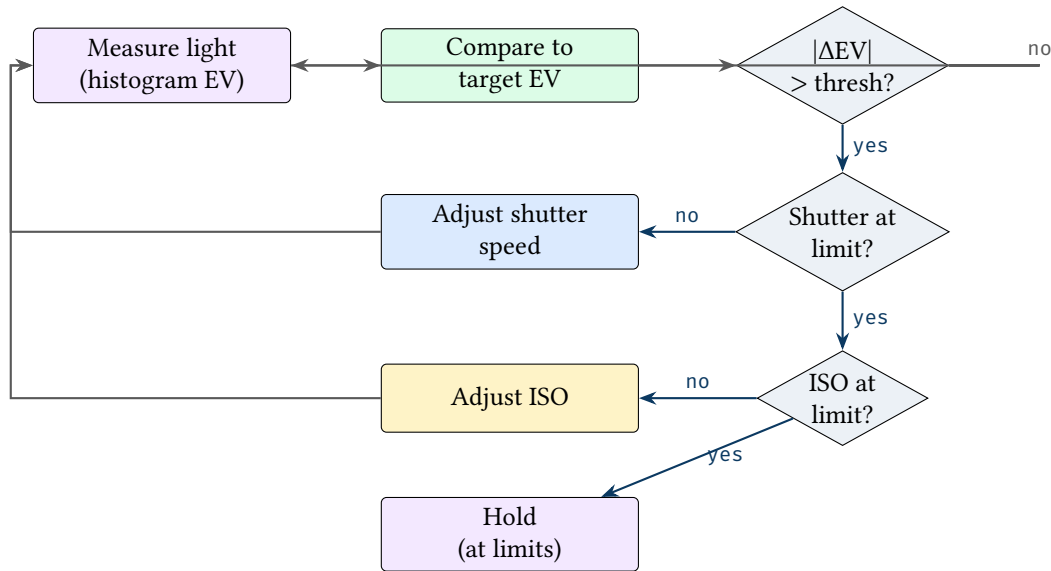


Figure 14. Automated exposure ramp strategy: measure light, compare to target, adjust shutter first, then ISO. Aperture is held constant throughout.

: Exposure Ramp Priority

Prioritize shutter speed adjustment over ISO to preserve signal-to-noise ratio; hold aperture constant to maintain optical quality and depth of field. Only increase ISO when shutter speed reaches its useful limit for the current target.

7.1 Exposure Lengths at Each Stage

The following table provides worked examples of specific exposure parameters for each eclipse stage. These values assume a full-frame sensor (Sony Alpha series) and serve as starting points; EclipsePhoto’s histogram guard rails will fine-tune within each stage.

Stage	Shutter	ISO	Aperture	Target
Partial (filtered)	1/1000s	100	f/8	Sunspot detail through solar filter
Diamond ring	1/4000s	100	f/5.6	Brilliant point; avoid blooming
Inner corona	1/500s	200	f/5.6	Coronal loops and base structure
Mid corona	1/60s	400	f/5.6	Streamer detail and helmet structure
Outer corona	1/2s	800	f/5.6	Faint extensions beyond $2 R_{\odot}$
Earthshine	2s	1600	f/5.6	Lunar surface illuminated by Earth
Shadow bands	1/500s	1600	f/2.8	Ground pattern (wide-angle rig)

Table 2. Worked exposure examples for each eclipse stage. These are starting points; EclipsePhoto adjusts based on histogram feedback.

7.2 Focus Verification Procedures

Achieving and maintaining critical focus is essential for eclipse photography, especially at telephoto focal lengths where depth of field is measured in fractions of a millimetre. The focus verification procedure should be completed well before C1 and the focus ring locked for the duration of the eclipse.

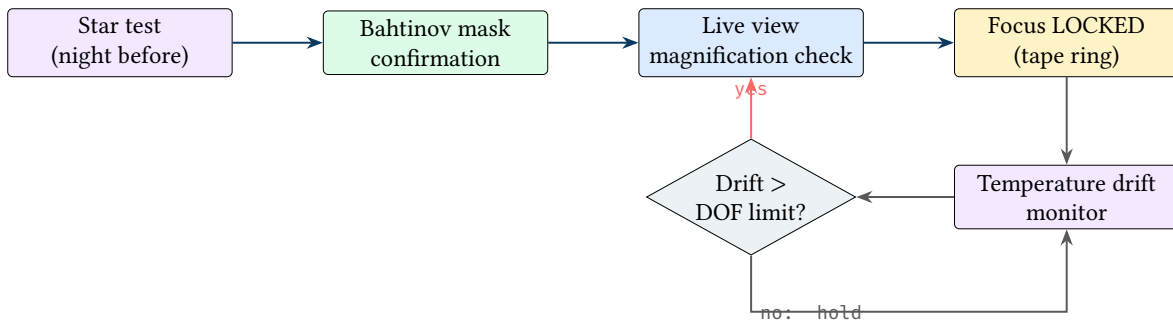
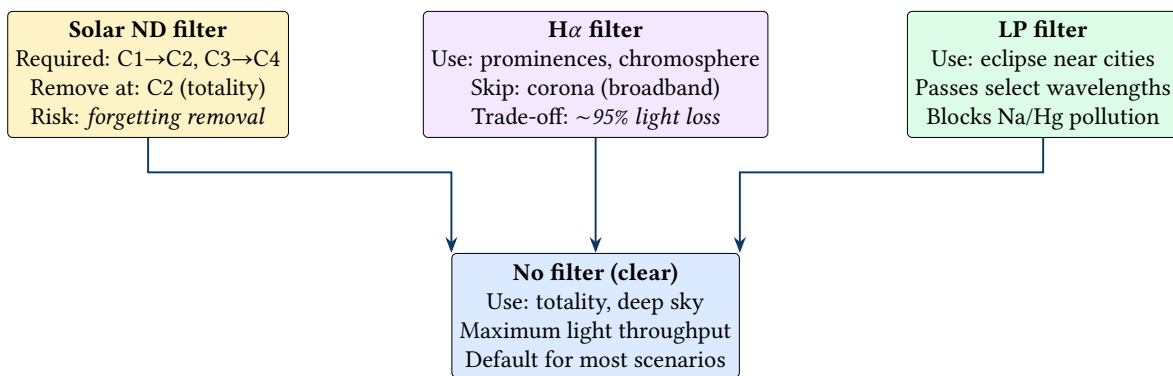


Figure 15. Focus verification procedure: star test establishes baseline, Bahtinov mask confirms critical focus, live view magnification provides final verification, then focus is locked. Temperature drift is monitored throughout the eclipse with re-check triggered if drift exceeds depth-of-field limits.

: Focus Lock Before C1

Lock focus before C1 and do not touch the focus ring during the eclipse. Any accidental bump to the focus ring during totality—when the operator is excited and potentially clumsy—will ruin every subsequent frame. Use gaffer tape to secure the focus ring after final verification.

8 Solar Filter Management



Filter removal is a critical automated or operator action—
EclipsePhoto should track filter state and warn if inappropriate

Figure 16. Filter selection logic: each filter serves a specific phase or target. Solar filter removal at C2 is the single most time-critical manual action during an eclipse.

8.1 Filter Removal Timing

The timing of solar filter removal around C2 is the most safety-critical moment in the entire eclipse sequence. Remove too early and the camera sensor (and the operator's eyes, if looking through an optical viewfinder) is exposed to dangerous levels of unfiltered sunlight. Remove too late and the diamond ring, chromosphere, and initial corona frames are lost behind the opaque filter.

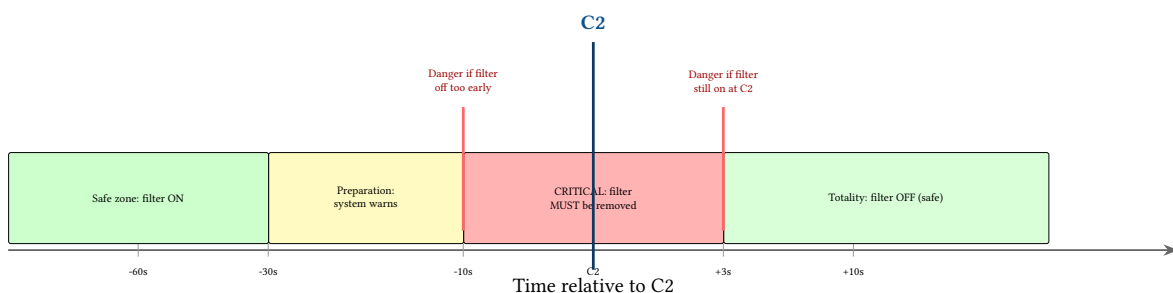


Figure 17. Filter removal timing around C2: the safe zone (green) transitions through a preparation warning zone (yellow) to the critical window (red) where the filter must be removed within seconds of C2. Removing too early risks sensor damage; too late wastes the diamond ring and chromosphere.

Sensor Damage Risk

Unfiltered sunlight through a telephoto lens can destroy a camera sensor in seconds. At 200 mm f/5.6, the lens concentrates approximately 10 W of solar energy onto the sensor—equivalent to a soldering iron. The solar filter must remain in place until the last possible moment before C2, and must be replaced immediately after C3. EclipsePhoto must track filter state and provide unmistakable audible and visual warnings at both transitions.

8.2 Safety Interlocks

EclipsePhoto implements multiple layers of safety to prevent filter-related errors:

- **Audible alarm:** A distinct audio tone begins 30 seconds before C2 (“prepare to remove filter”) and changes to an urgent pattern at C2 (“remove filter NOW”). A similar sequence occurs at C3 (“replace filter NOW”).
- **Full-screen warning:** The iPhone hub displays a full-screen overlay during the filter transition windows: green background when filter state matches the expected state, red background with flashing border when action is required.
- **Operator confirmation:** The system requires explicit confirmation (button press or voice command) before transitioning from FILTER_REMOVE to TOTALITY state, and from FILTER_REPLACE to PARTIAL_EXIT state. This prevents the automation from proceeding if the operator has not actually removed or replaced the physical filter.

Filter State Tracking

FILTER_REMOVE and FILTER_REPLACE are **operator-confirmed** states. The system should: (1) sound an audible alarm, (2) display a full-screen warning, (3) require explicit confirmation before proceeding to TOTALITY or PARTIAL_EXIT. The system cannot physically verify filter presence—it relies on the operator to confirm.

9 Multi-Camera Deployment and Rig Strategy

An EclipsePhoto deployment typically uses multiple camera nodes for complementary coverage. The optimal geometry depends on the target: parallax-based 3D reconstruction requires baseline separation, panorama stitching requires overlapping fields, and multi-focal-length setups require co-located but differently configured nodes.

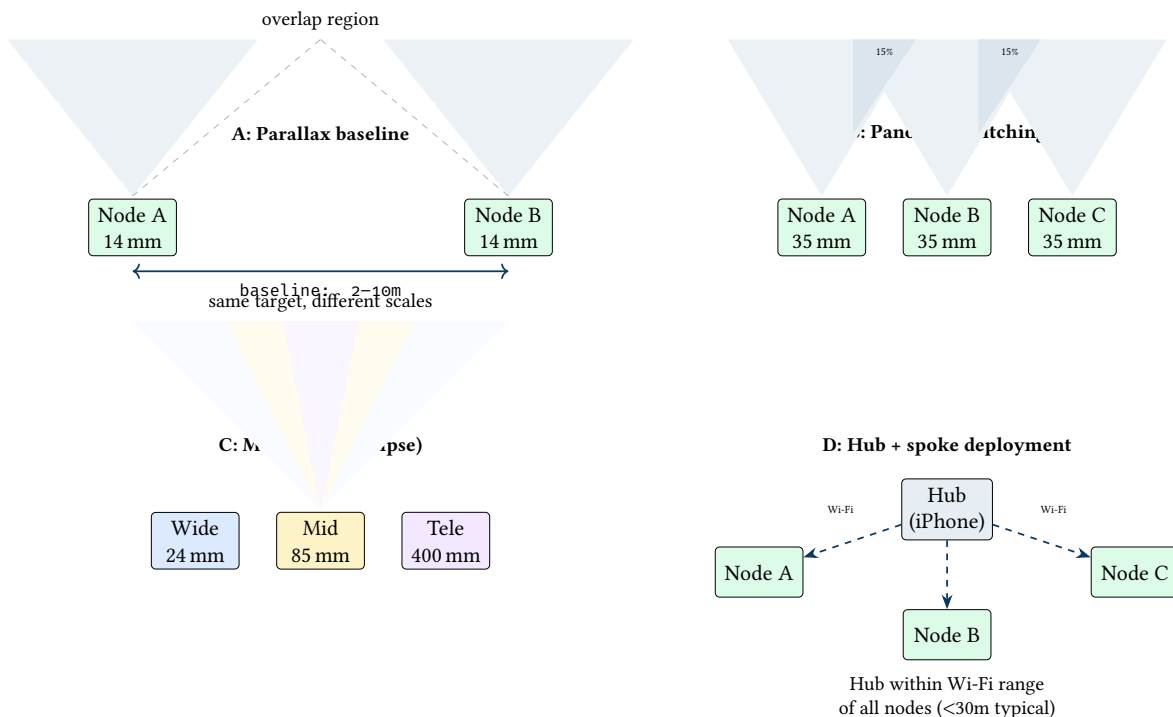


Figure 18. Multi-camera deployment geometries: (A) parallax baseline for potential 3D reconstruction, (B) panorama stitching with ~15% overlap, (C) co-located multi-focal-length for simultaneous wide/narrow coverage, (D) hub-and-spoke Wi-Fi topology constraint.

A complete eclipse imaging setup typically requires two or three rigs with different focal lengths and tracking strategies. The key insight is that the corona extends far beyond the solar disc—up to 3+ solar radii ($\sim 3^\circ$ total diameter)—so a telephoto rig zoomed tightly on the Sun will miss the outer corona. Simultaneously, a wide-angle rig captures the eclipse in its landscape context for timelapse sequences.

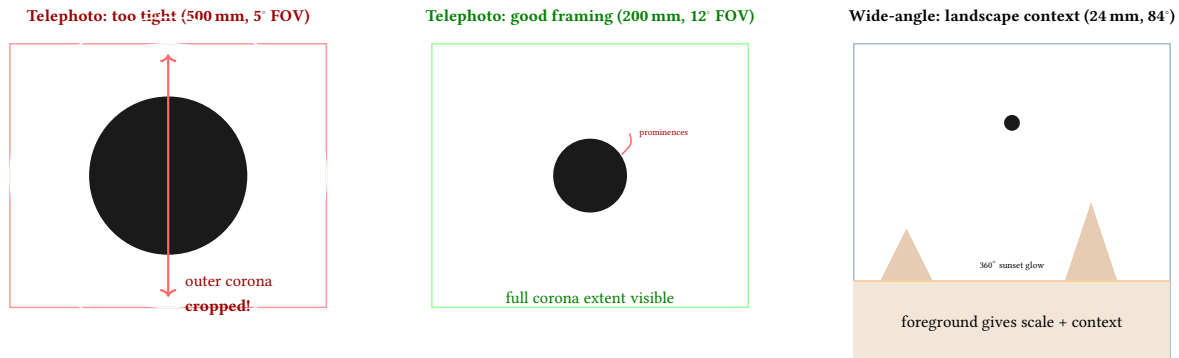


Figure 19. Corona framing: a 500 mm telephoto (left) crops the outer corona; 200 mm (center) provides room for the full coronal extent including streamers; 24 mm wide-angle (right) captures the eclipse in landscape context with the 360° horizon glow visible during totality.

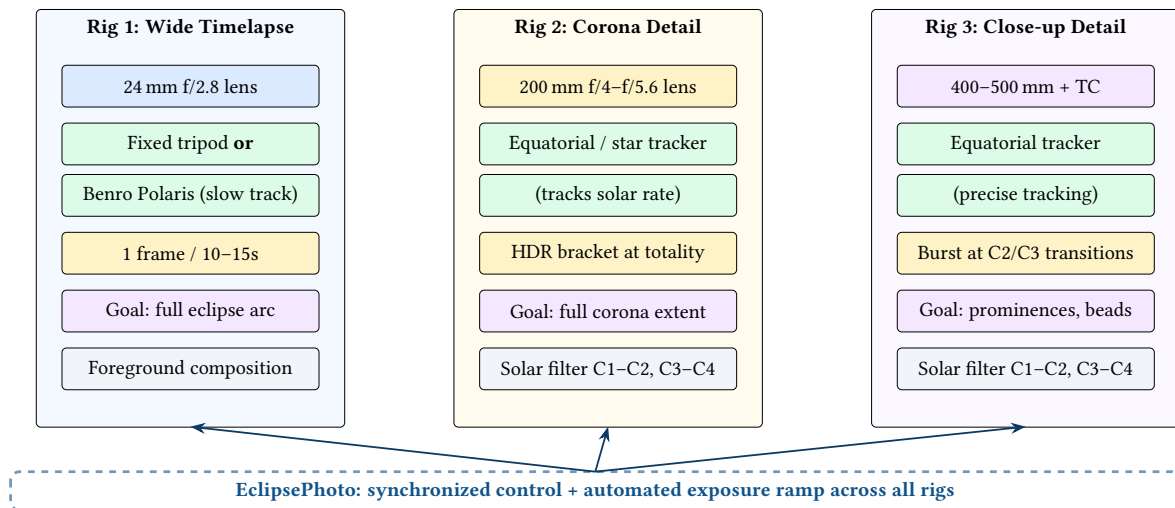


Figure 20. Three-rig eclipse deployment: (1) wide-angle on tripod or Benro Polaris for timelapse with foreground, (2) tracked 200 mm for full corona HDR, (3) tracked 400–500 mm for close-up prominences and transient events. EclipsePhoto coordinates exposure timing across all three.

The wide-angle rig has two operational modes:

- **Stationary (tripod):** Fixed composition with foreground (landmark, people, horizon). The Sun moves through the frame over the eclipse duration, creating a composite arc showing the progression from crescent to totality and back. Requires planning the Sun's path to ensure it stays in frame for the full 2–3 hours.
- **Slow-tracked (Benro Polaris):** The mount rotates slowly to follow the Sun, keeping it centered while the foreground gradually shifts. This produces smoother timelapse sequences and avoids the Sun drifting out of a tighter composition, but sacrifices the dramatic composite arc. Best for video-oriented output or when using moderate focal lengths (35–50 mm) where the Sun's motion would otherwise exit the frame.

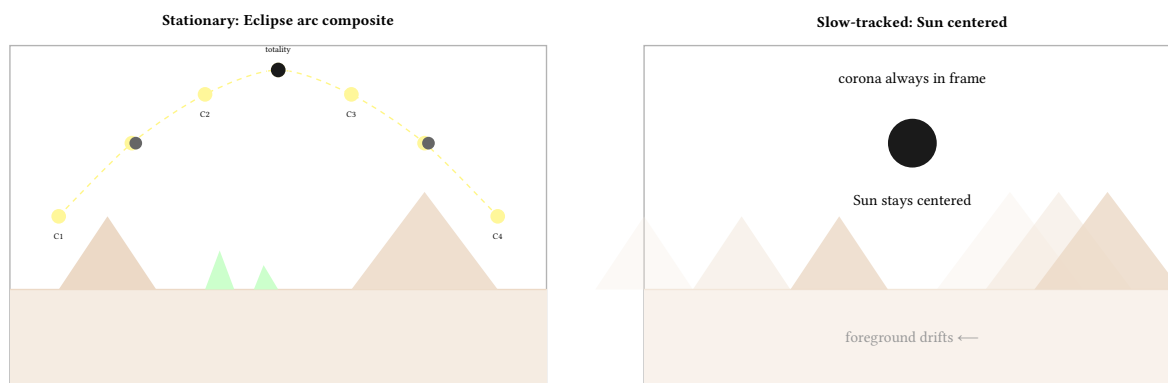


Figure 21. Wide-angle timelapse strategies: stationary (left) produces a dramatic composite arc showing the eclipse progression against fixed foreground—the classic “eclipse arc” image; slow-tracked on a Benro Polaris (right) keeps the Sun centered for smoother video timelapse but the foreground drifts. Both require planning the composition for the full eclipse duration.

Design Decision 1: Corona Framing Rule

For corona photography, select a focal length that provides at least 3° of sky around the eclipsed Sun (6° total field minimum). At 200 mm on full-frame (12° diagonal FOV), the corona fills roughly half the frame—ideal. At 500 mm (5° FOV), the outer corona and any extended streamers will be cropped. When in doubt, frame wider: you can crop in post-processing but cannot recover cropped corona.

9.1 Rig Coordination via EclipsePhoto

The iPhone hub synchronizes exposure timing, filter state, and bracket sequences across all three rigs. The coordination model differs by rig role:

- **Wide rig (Rig 1):** Runs independently in timelapse mode throughout the eclipse. The hub sets the initial interval and exposure parameters but does not send per-frame commands. The wide rig does not use a solar filter (it shoots at wide angle where the Sun is a small point) and does not require filter-state synchronization.
- **Corona rig (Rig 2) and Close-up rig (Rig 3):** Receive synchronized C2/C3 commands from the hub. At C2, both rigs receive the “REMOVE FILTER” warning simultaneously, followed by the “BEGIN TOTALITY” command that starts their respective bracket or burst sequences. At C3, both receive “REPLACE FILTER” in lockstep.

The hub uses GPS-synchronized time to ensure that bracket start commands arrive at each node within 100 ms of each other, preventing one rig’s long exposure from vibrating the tripod during another rig’s short exposure (if co-mounted).

10 Focal Length and Field of View

Target selection drives focal length choice. The solar corona extends $\sim 3^\circ$ from the Sun’s center, and deep-sky objects vary widely. A field-of-view diagram helps match lenses to targets and informs the 500-rule calculations in the shutter speed subsection.

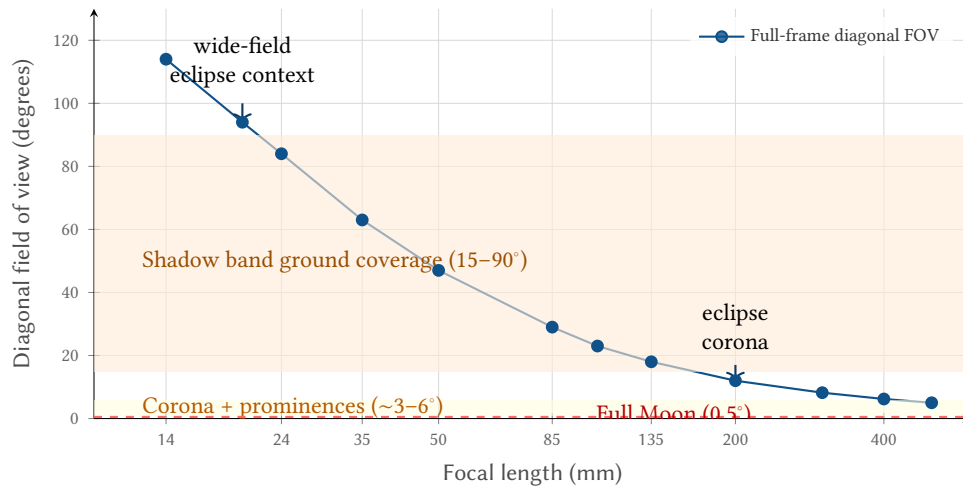


Figure 22. Focal length vs. diagonal field of view (full-frame sensor) with target angular sizes overlaid. Wide-angle lenses (14–35 mm) suit landscape context and shadow band capture; telephoto (200–500 mm) is needed for corona detail and lunar surface.

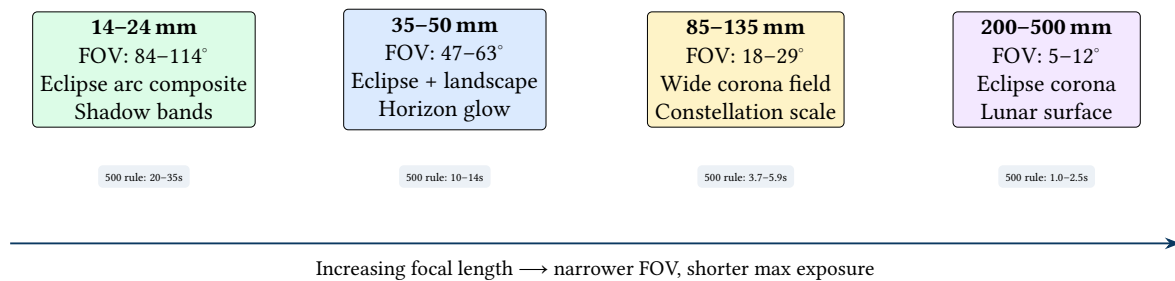


Figure 23. Lens focal length categories with target assignments and maximum untracked exposure times (500 rule). Longer focal lengths demand faster shutter speeds, compressing the available exposure range.

11 HDR Compositing Pipeline

The 12+ EV dynamic range of the solar corona requires compositing multiple exposures. The processing pipeline aligns frames, weights them by exposure, and applies specialized radial filtering to reveal coronal structure that no single exposure can capture.

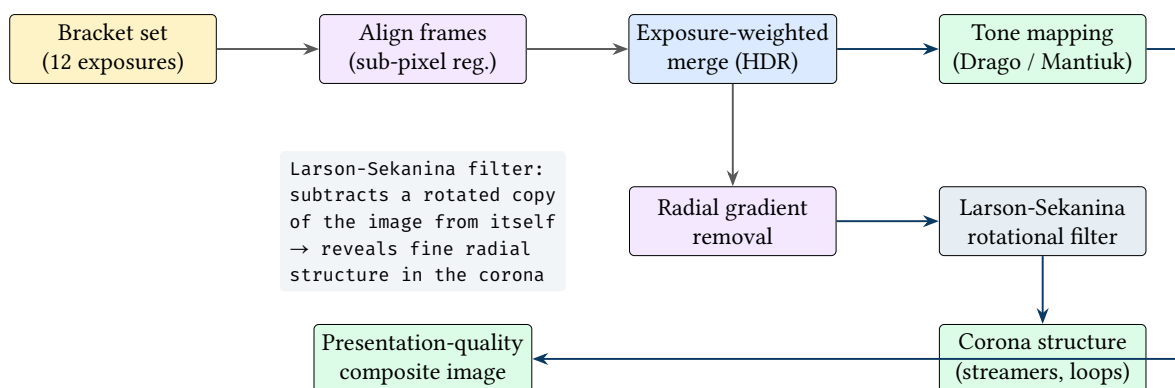


Figure 24. HDR compositing pipeline for corona imagery: bracket exposures are aligned, merged with exposure-based weighting, and processed through two paths—tone mapping for natural appearance, and radial filtering (Larson-Sekanina) to reveal fine coronal structure.

11.1 Noise Reduction via Stacking

During totality, multiple bracket sets are captured in sequence. If totality lasts 4 minutes and a full 12-exposure bracket takes approximately 30 seconds (including readout and settling time), approximately 8 bracket sets can be acquired. Each bracket set contributes one frame at each exposure level.

For the faintest exposures (outer corona at 1–2 seconds), individual frames are dominated by read noise and photon shot noise. Stacking N aligned frames from successive bracket sets improves the signal-to-

noise ratio by a factor of $\sqrt{N}$. With 8 bracket sets, stacking the outer-corona frames yields $\sqrt{8} \approx 2.8\times$ SNR improvement—equivalent to roughly 1.5 stops of additional effective dynamic range at the faint end.

The alignment step is straightforward for tracked rigs (the Sun remains stationary in the frame), but for untracked rigs the solar disc drifts by approximately 15 arcseconds per second. Over a 4-minute totality, this amounts to $\sim 1^\circ$ of drift—significant at telephoto focal lengths. EclipsePhoto should record timestamps with each frame to enable precise alignment during post-processing.

12 Pre-Totality Operational Timeline

An eclipse totality sequence requires precisely timed actions. This operational countdown ties the physical phenomena (contact points, illuminance changes) to the concrete camera operations that EclipsePhoto must automate or prompt.

Time	Action
T–24h	Check solar activity (SDO, SWPC); note active regions for prominence targets
T–4h	Equipment setup; GPS sync verification; test exposures on daytime sky
T–2h	Star focus calibration (if pre-dawn setup); Bahtinov mask on bright star
T–1h	Solar filter on all rigs; frame composition; test filtered exposure on Sun
T–60min	C1: begin partial-phase capture (1 frame / 2 min)
T–30min	Verify tracking; check focus has not drifted from thermal changes
T–10min	Temperature drop check; arm bracket sequence on corona and close-up rigs
T–5min	Final focus verification via live view magnification; prepare totality exposure table
T–30s	Shadow bands may appear; switch wide rig to video mode if desired
T=0 (C2)	REMOVE SOLAR FILTER ; start HDR bracket on all telephoto rigs
T+mid	Execute brackets continuously; check histogram after each set; adjust if needed
T=end (C3)	REPLACE SOLAR FILTER ; resume partial-phase capture
T+60min	C4: end of eclipse; stop all capture sequences

Table 3. Expanded operational timeline from T–24h through C4. Bold entries are safety-critical filter actions.

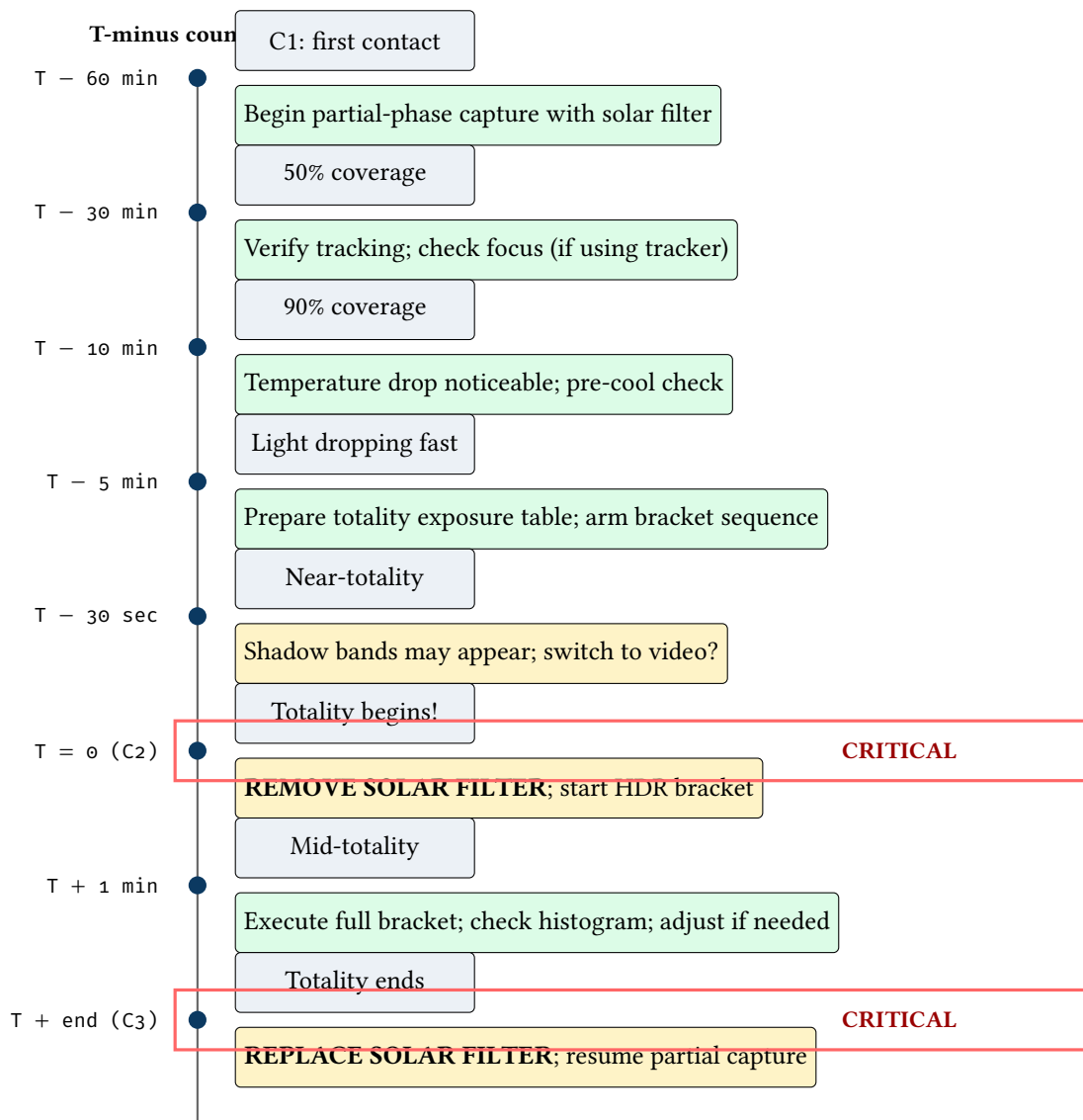


Figure 25. Pre-totality operational countdown: timed sequence of camera actions keyed to eclipse contact points. Solar filter removal at C2 and replacement at C3 are the most safety-critical actions (both for equipment and for image capture).

Solar Filter Safety

Failure to remove the solar filter at C2 wastes the entire totality window (the filter blocks >99.999% of light). Failure to replace the filter at C3 risks sensor damage from unfiltered sunlight. EclipsePhoto should track filter state and issue prominent warnings at C2 and C3 transitions.

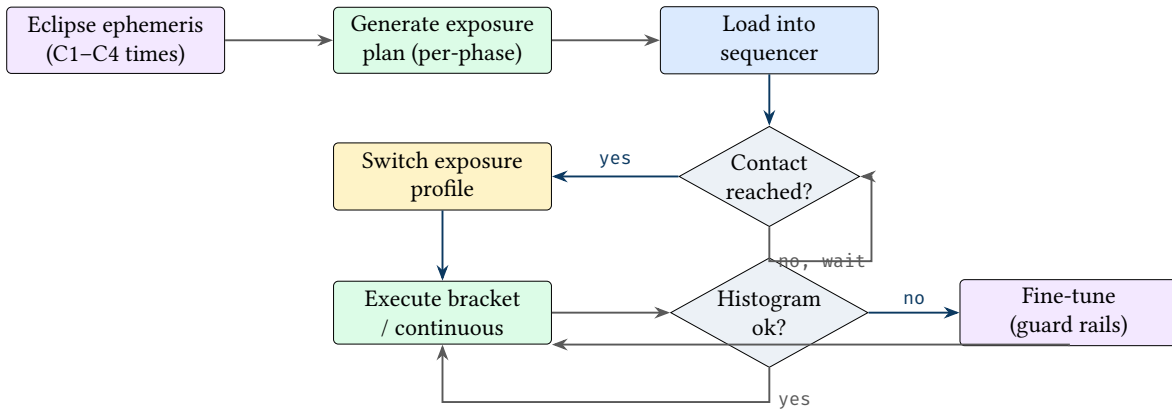
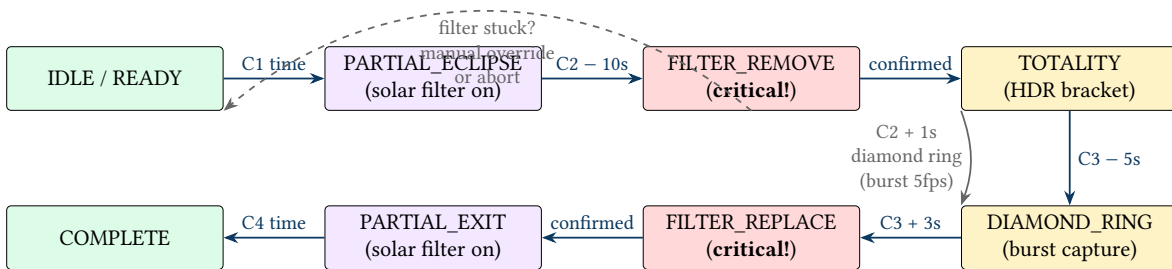


Figure 26. Eclipse automation pipeline: pre-computed ephemeris drives a time-keyed exposure plan, with histogram-based guard rails providing fine-tuning within each phase. This combines the predictable timing of eclipses with reactive histogram-based guard rails.

13 Eclipse State Machine

The capture node state machine defined in the system architecture can be extended with eclipse-specific states that map to the physical phenomena described in this report. This bridges the gap between the astronomical science and the system architecture.



FILTER_REMOVE and FILTER_REPLACE are **operator-confirmed** states.
The system should: (1) sound an audible alarm, (2) display a full-screen warning,
(3) require explicit confirmation before proceeding to TOTALITY or PARTIAL_EXIT.

Figure 27. Eclipse-specific state machine: extends the capture node states with eclipse phases. FILTER_REMOVE (at C2) and FILTER_REPLACE (at C3) are operator-confirmed critical transitions. The DIAMOND_RING state triggers burst capture at the C2/C3 transitions.

14 Telemetry Dashboard

During an eclipse, the iPhone hub must display time-critical information at a glance. This mockup shows the essential telemetry elements for eclipse operations, organized by urgency.

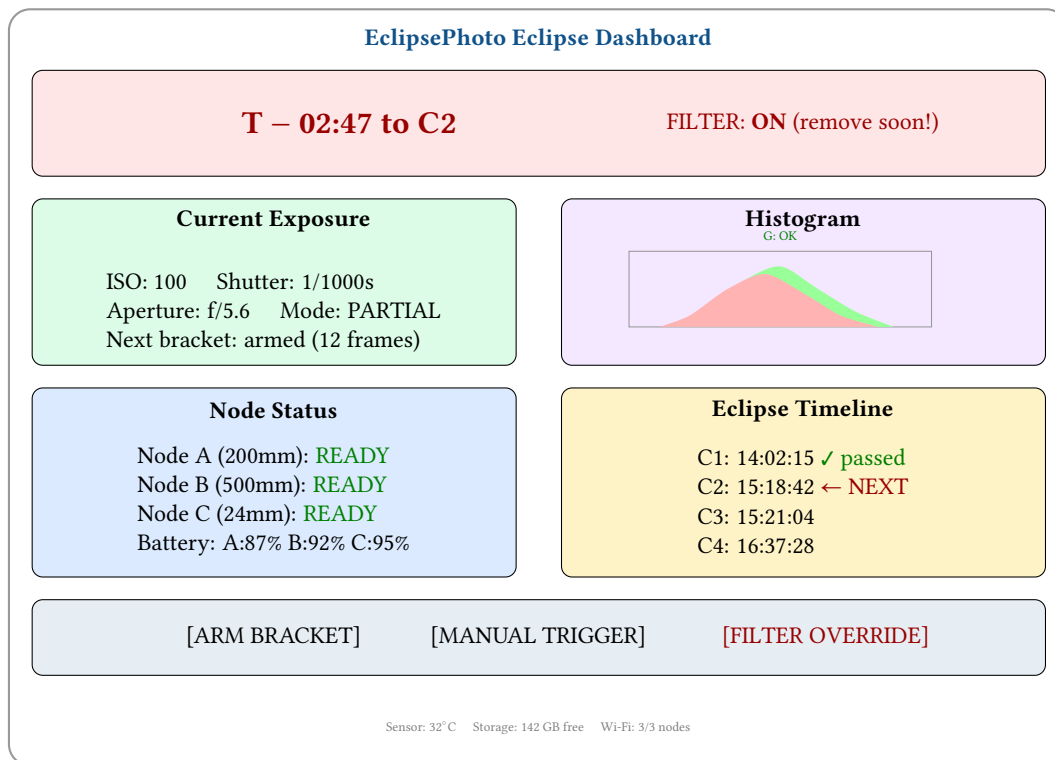


Figure 28. Eclipse telemetry dashboard mockup: the countdown to the next contact point dominates the top bar; filter state is always visible with prominent warnings. Current exposure, live histogram, node status, and eclipse timeline are shown in a four-panel layout with action buttons for manual override.

Design Decision 2: Dashboard Design: Eclipse Mode

The eclipse dashboard prioritizes three things above all else: (1) countdown to next contact point, (2) solar filter state with audible/visual warnings, and (3) bracket sequence readiness. All other telemetry (node health, storage, temperature) is secondary. The filter state indicator must be the largest, most prominent element on screen during the C2 and C3 transitions.

15 Focus and Optical Quality

15.1 Star Focus Calibration

Critical focus at telephoto focal lengths requires calibration on a point source—ideally a bright star. The night before the eclipse (or in the pre-dawn hours if the setup is done early), point the telephoto rig at a bright star (Vega, Arcturus, or Sirius work well) and use live view at maximum magnification to achieve the smallest possible point spread function.

The star test procedure is straightforward:

1. Point the telescope or telephoto lens at a bright star.
2. Switch to manual focus and enable live view at maximum magnification (10× or higher).
3. Slowly adjust the focus ring until the star image is the smallest, sharpest point. Overshoot in both directions to confirm you have found the true minimum.
4. If the camera supports focus peaking, enable it as a secondary confirmation.
5. Lock the focus ring with gaffer tape.

If the eclipse occurs during the day and no pre-dawn setup is possible, focus can be achieved on the solar disc itself through the solar filter, using sunspot edges or the solar limb as focus targets. This is less precise than a star test but adequate for most focal lengths below 400 mm.

15.2 Temperature-Induced Focus Drift

Telescope and telephoto lens barrels are typically constructed from aluminium alloys, which expand and contract with temperature changes. During an eclipse, the ambient temperature can drop by 3–8°C in the final minutes before totality as the Sun's energy input decreases. This thermal contraction shifts the focal plane.

At 200 mm f/4, a 10°C temperature drop can shift focus by approximately 50–100 μm . For context, the depth of field at f/5.6 and 200 mm focused at infinity is roughly 200 μm —so a 10°C swing can consume half the available depth of field.

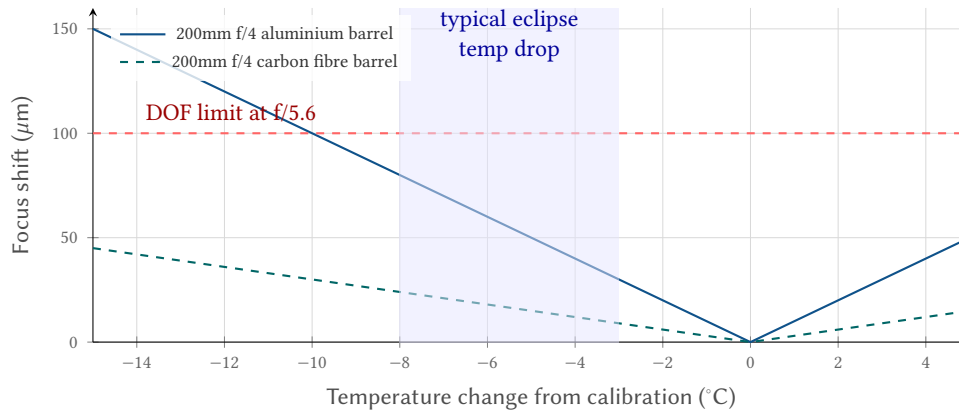


Figure 29. Focus shift vs. temperature change for typical telephoto lenses. The blue shaded region shows the 3–8°C temperature drop typical during the final partial phases of an eclipse. Aluminium-barrel lenses can shift by 30–80 μm in this range, approaching the depth-of-field limit at f/5.6.

The system should log ambient temperature (from the Raspberry Pi’s onboard sensor or an external probe) and warn if the cumulative temperature change since focus calibration exceeds a configurable threshold (default: 5°C). If pre-compensation is desired, the operator can intentionally defocus slightly in the direction of expected drift.

15.3 Bahtinov Mask Procedure

A Bahtinov mask is an optical device placed over the lens aperture that creates a distinctive three-spoke diffraction pattern from point sources. Perfect focus is achieved when the central spoke exactly bisects the angle formed by the other two spikes. Any defocus causes the central spoke to shift laterally, providing a clear and unambiguous visual indicator.

The Bahtinov mask is the gold standard for focus confirmation because it removes the subjectivity of judging “minimum star size” by eye. For telephoto lenses, commercial Bahtinov masks are available in standard filter thread sizes; for telescopes, they can be 3D-printed or laser-cut to fit the aperture.

15.4 Lens Breathing

Some lenses—particularly cinema-style lenses and certain older designs—exhibit “focus breathing”: a slight change in field of view when the focus position changes. This means that after achieving perfect focus, the framing may have shifted slightly compared to the initial composition.

Always verify framing *after* the final focus adjustment, not before. If the lens breathes significantly (more than 2–3% FOV change), re-compose after locking focus and note that the framing will be stable from that point forward.

: Focus Lock Protocol

Lock focus before C1. Do not adjust focus during the eclipse. If temperature drift is expected, pre-compensate based on the forecast temperature at mid-totality. Verify framing after locking focus to account for any lens breathing.

16 Robustness and Sanity Checks

16.1 Pre-Event Dry Run

On the day before the eclipse (or earlier), run the full EclipsePhoto automation sequence with simulated contact times set 5 minutes apart. This dry run should exercise every state transition:

1. IDLE → PARTIAL_ECLIPSE: verify the intervalometer starts and captures filtered frames.
2. PARTIAL_ECLIPSE → FILTER_REMOVE: verify the audible alarm sounds and the full-screen warning appears.
3. FILTER_REMOVE → TOTALITY: verify the bracket sequence starts after operator confirmation.
4. TOTALITY → DIAMOND_RING: verify the burst capture triggers at the correct time offset.
5. DIAMOND_RING → FILTER_REPLACE: verify the replacement warning sounds.

6. FILTER_REPLACE → PARTIAL_EXIT: verify filtered capture resumes.
7. PARTIAL_EXIT → COMPLETE: verify clean shutdown.

Any failures in the dry run must be resolved before the actual eclipse. There is no opportunity to debug during totality.

16.2 Exposure Verification

After each exposure, EclipsePhoto should perform a rapid histogram analysis to verify that the image is neither clipped nor underexposed. The guard rails are:

- **Highlight clipping:** If the 99.5th percentile of pixel values exceeds 252/255 (in 8-bit representation), the frame is considered clipped. Reduce exposure by 1 stop for the next frame and flag the current frame.
- **Shadow noise floor:** If the 0.5th percentile of pixel values is below 3/255, the frame may be underexposed (or the scene is simply very dark, as expected for outer corona frames). If the current stage expects bright content (inner corona, diamond ring), increase exposure by 1 stop.

These guard rails operate within each stage's expected range and do not override the bracket sequence—they adjust the baseline around which the bracket is centered.

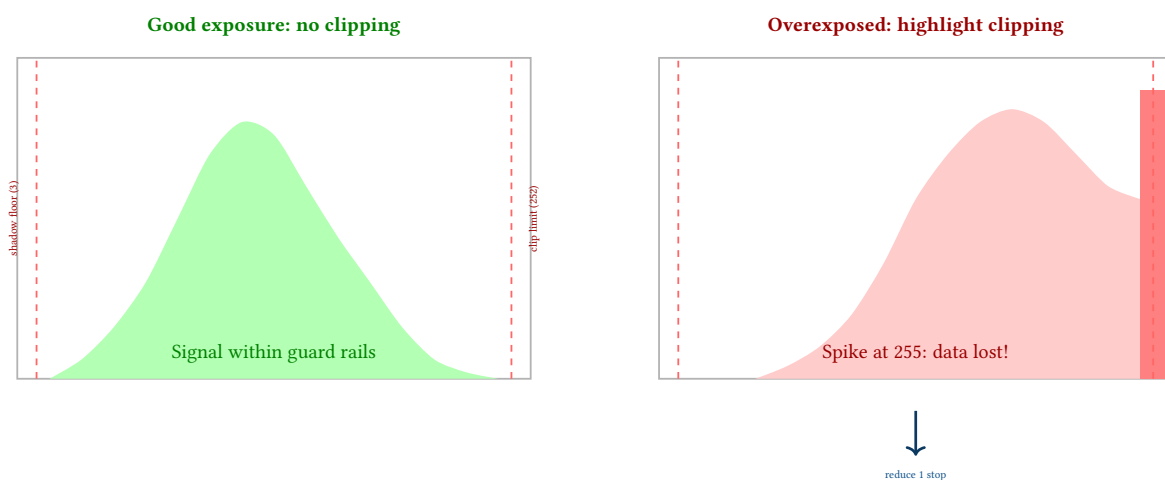


Figure 30. Histogram guard rails: a well-exposed frame (left) has signal between the shadow floor (3/255) and clip limit (252/255). A clipped frame (right) shows a spike at 255—irrecoverable data loss in highlights. EclipsePhoto should detect clipping and adjust exposure by 1 stop for the next frame.

16.3 Storage and Battery Margins

Eclipse photography generates a large volume of data in a short time, especially during the HDR bracket sequences of totality. The following table estimates storage requirements per rig:

Item	Data	Cumulative
12-frame bracket at 50 MB/frame	600 MB	600 MB
8 brackets during totality	×8	4.8 GB
Partial phase: 1 frame/2 min × 120 min	60 frames	3.0 GB
Burst captures at C2/C3 (50 frames each)	100 frames	5.0 GB
Total per rig (minimum)		~13 GB
Recommended card size		64 GB+

Table 4. Storage requirements per rig. Recommend 64 GB+ cards with a safety margin of 5× to allow for additional opportunistic captures.

Battery margins should target at least 150% of the estimated shot count. A typical mirrorless camera battery provides 400–500 shots (CIPA rating); the eclipse sequence may require 300–400 shots per rig. Carry a spare battery per rig and verify charge levels at T–4h. If using battery grips with dual batteries, ensure both slots are charged.

16.4 Thermal Management

Cameras and Raspberry Pi nodes operating in direct sunlight can reach temperatures exceeding 45°C, especially in the 60–90 minutes of partial phase before totality. At these temperatures, sensor noise increases, the Raspberry Pi may throttle its CPU, and some cameras will display overheating warnings or shut down entirely.

Mitigation strategies include:

- **Reflective covers:** Use white or silver reflective material (space blankets, white foam board) to shade the camera body and Pi enclosure from direct sunlight.
- **Minimise live view:** Keep the camera's rear LCD off when not actively composing. Live view significantly increases sensor heating.
- **Monitor CPU temperature:** EclipsePhoto should read the Pi's thermal zone and warn if the CPU temperature exceeds 70°C (throttling typically begins at 80°C).
- **Shade structures:** If possible, set up a canopy or umbrella over the equipment. This also helps with screen visibility in bright sunlight.

Thermal Shutdown Risk

A camera that overheats and shuts down during totality cannot be recovered in time. The 2–7 minute totality window is too short to cool the camera, restart it, and reconfigure exposure settings. Thermal management must be addressed *before* C1, not during the event. Test the full equipment stack in direct sunlight for at least 90 minutes during the dry run.

Revision History

Version	Date	Changes
0.1.0	March 2026	Initial technical report; eclipse science + operational design